# Measurable Selections of Multifunctions

## Chapter 10c: Applications of Fixed Point Theorems for Multifunctions

Based on Pathak's *An Introduction to Nonlinear Analysis and Fixed Point Theory*

# Outline

# Measure Spaces and Measurability

## Definition (Measure Space)

A **measure space** is a triple $(\Omega, \mathcal{A}, \mu)$ where:

- $\Omega$ is a nonempty set (sample space)
- $\mathcal{A}$ is a $\sigma$-algebra of subsets of $\Omega$
- $\mu : \mathcal{A} \to [0, \infty]$ is a **measure** (countably additive function)

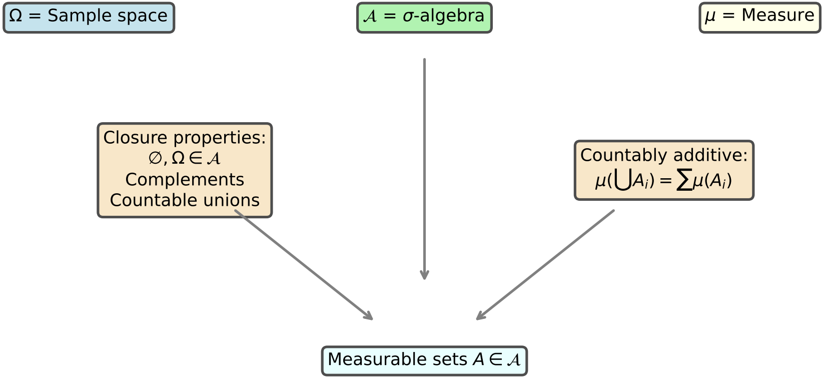
## Definition ($\sigma$-algebra)

A collection $\mathcal{A}$ of subsets of $\Omega$ is a $\sigma$-algebra if:

1. $\emptyset, \Omega \in \mathcal{A}$
2. If $A \in \mathcal{A}$, then $A^c \in \mathcal{A}$ (closed under complements)
3. If $\{A_n\}_{n=1}^{\infty} \subseteq \mathcal{A}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$ (closed under countable unions)

# Measure Space Structure

**Measure Space:** $(\Omega, \mathcal{A}, \mu)$



# Measurable Functions

## Definition (Measurable Function)

Let $(\Omega, \mathcal{A})$ be a measurable space and $(Y, \mathcal{V})$ be a topological space. A mapping $f : \Omega \to Y$ is **measurable** if

$$f^{-1}(V) \in \mathcal{A} \quad \text{for every open set } V \subseteq Y.$$

## Theorem (Properties of Measurable Functions)

*If $f, g : \Omega \to \mathbb{R}$ are measurable functions, then:*

1. $f + g$, $f - g$, $f \cdot g$ *are measurable*
2. $|f|$, $f^+$, $f^-$ *are measurable*
3. $\sup_n f_n$, $\inf_n f_n$, $\limsup_n f_n$ *are measurable (if limits exist)*

# Simple Functions

## Definition (Simple Function)

A function $s : \Omega \to [0, \infty)$ is **simple** if its range is a finite set. If $c_1, c_2, \ldots, c_n$ are the distinct values and

$$E_i = \{x \in \Omega : s(x) = c_i\}, \quad i = 1, 2, \ldots, n,$$

then

$$s(x) = \sum_{i=1}^{n} c_i \chi_{E_i}(x),$$

where $\chi_{E_i}$ is the characteristic function of $E_i$.

## Theorem (Approximation by Simple Functions)

*Let $u : \Omega \to [0, \infty)$ be a measurable function. There exists a sequence $\{u_n\}$ of simple functions such that:*

1. $0 \leq u_1(x) \leq u_2(x) \leq \cdots \leq u(x)$ *for all* $x \in \Omega$
2. $u_n(x) \to u(x)$ *as* $n \to \infty$ *for all* $x \in \Omega$

# Multifunctions: Definition and Basics

## Definition (Multifunction)

Let $X$ and $Y$ be nonempty sets. A **multifunction** (or set-valued map) is a mapping

$$F : X \to \mathcal{P}(Y),$$

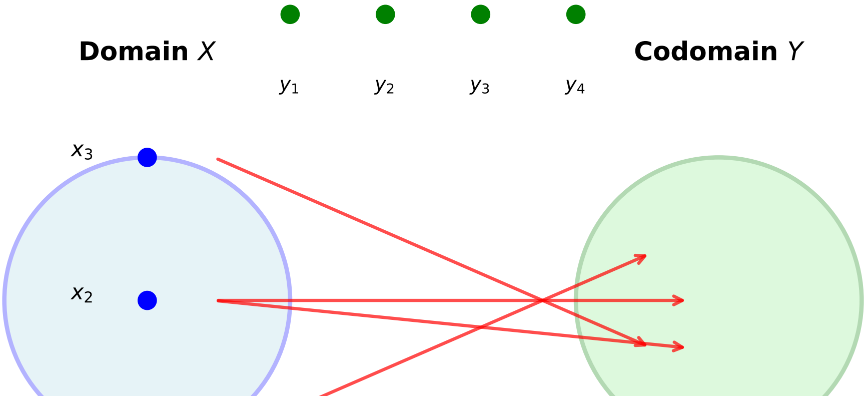
where $\mathcal{P}(Y)$ denotes the power set (family of all subsets of $Y$).
For each $x \in X$, the image $F(x) \subseteq Y$ is the set of all possible values associated with $x$.

## Example

- $F : \mathbb{R} \to \mathcal{P}(\mathbb{R})$ defined by $F(x) = \{y \in \mathbb{R} : y^2 = x\}$
- Correspondence in economics: price set $\to$ demand bundles
- Differential inclusion: $x'(t) \in F(t, x(t))$

# Multifunction Illustration

**Multifunction:** $F : X \to \mathcal{P}(Y)$



# Types of Multifunctions

## Definition

A multifunction $F : X \to \mathcal{P}(Y)$ is said to have:

- **Closed values** if $F(x)$ is a closed set for every $x \in X$
- **Convex values** if $F(x)$ is a convex set for every $x \in X$
- **Compact values** if $F(x)$ is a compact set for every $x \in X$
- **Nonempty values** if $F(x) \neq \emptyset$ for every $x \in X$

## Definition (Graph of a Multifunction)

The **graph** of $F : X \to \mathcal{P}(Y)$ is defined as

$$\mathrm{Graph}(F) = \{(x, y) \in X \times Y : y \in F(x)\}.$$

# Hausdorff Distance

## Definition (Hausdorff Distance)

For nonempty subsets $A, B$ of a metric space $(Y, d)$, the **Hausdorff distance** is

$$H(A, B) = \max\left\{\sup_{a\in A} d(a, B), \sup_{b\in B} d(b, A)\right\},$$

where $d(a, B) = \inf_{b\in B} d(a, b)$.

- $H$ defines a metric on the space of nonempty compact sets
- Useful for measuring the distance between image sets
- Essential for studying continuity of multifunctions

# Selection Functions

## Definition (Selection)

Let $F : X \to \mathcal{P}(Y)$ be a multifunction. A function $f : X \to Y$ is called a **selection** (or **selector**) of $F$ if

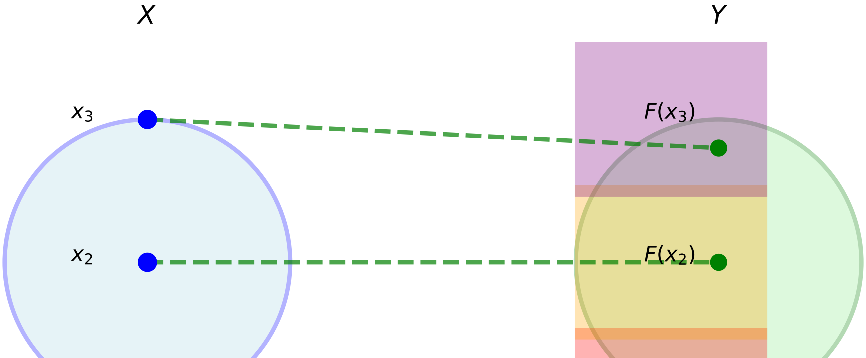
$$f(x) \in F(x) \quad \text{for all } x \in X.$$

## Example

For $F(x) = \{y \in \mathbb{R} : |y| \leq x\}$:

- $f_1(x) = x$ is a selection
- $f_2(x) = -x$ is a selection
- $f_3(x) = \frac{x}{2}$ is a selection

# Selection Function Illustration



# Measurable Selections

**Definition (Measurable Selection)**

Let $(\Omega, \mathcal{A}, \mu)$ be a measure space, $X$ a separable Banach space, and $F : \Omega \to \mathcal{P}(X)$ a multifunction. A measurable function $s : \Omega \to X$ is a **measurable selection** of $F$ if:

1. $s(\omega) \in F(\omega)$ for $\mu$-almost every $\omega \in \Omega$
2. $s$ is (Borel) measurable with respect to $\mathcal{A}$ and the Borel $\sigma$-algebra of $X$

Measurable selections are crucial for:

- Proving existence of solutions to integral and differential inclusions
- Establishing measurability properties of dependent choices
- Optimal control and variational problems

## Definition (Measurable Multifunction)

A multifunction $F : \Omega \to \mathcal{P}(Y)$ (with $\Omega$ a measure space and $Y$ a separable Banach space) is said to be:

- **Weakly measurable** if for each $y \in Y$, the function

$$\omega \mapsto d(y, F(\omega)) = \inf_{f \in F(\omega)} \|y - f\|$$

  is measurable.
- **Strongly measurable** if there exist measurable selections $s_n : \Omega \to Y$ such that

$$F(\omega) = \overline{\{s_n(\omega) : n \in \mathbb{N}\}} \text{ for a.e. } \omega.$$

# Measurability Conditions

## Theorem (Measurability Equivalent Conditions)

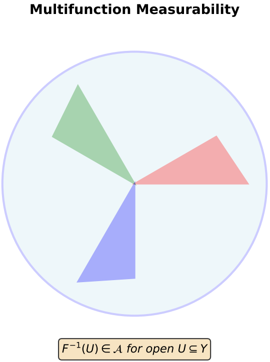
*For a multifunction $F : \Omega \to \mathcal{P}(Y)$ with $(\Omega, \mathcal{A}, \mu)$ a measure space and $Y$ a separable Banach space:*

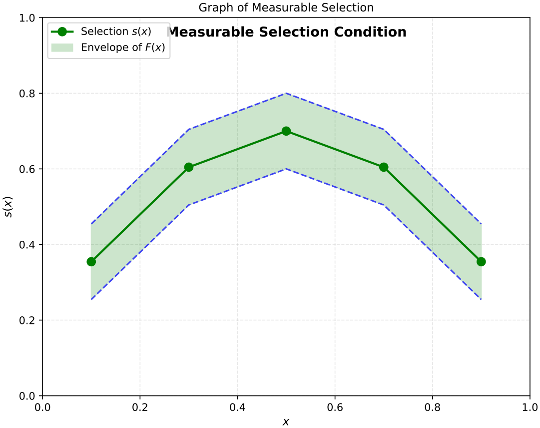
1. $F^{-1}(U) := \{\omega \in \Omega : F(\omega) \cap U \neq \emptyset\} \in \mathcal{A}$ *for all open* $U \subseteq Y$
2. *For each closed* $C \subseteq Y$*, we have* $F_c^{-1}(C) := \{\omega : F(\omega) \subseteq C\} \in \mathcal{A}$
3. $F$ *admits a countable family of measurable selections* $\{f_n\}$ *such that*

$$F(\omega) = \overline{\{f_n(\omega) : n \in \mathbb{N}\}} \text{ for a.e. } \omega \in \Omega.$$

*These conditions are equivalent when $F$ has closed, nonempty values.*

# Measurability Illustration



# Filipov's Selection Theorem

**Theorem (Filipov Selection Theorem)**

*Let $(\Omega, \mathcal{A}, \mu)$ be a $\sigma$-finite measure space, $X$ a separable Banach space, and $F : \Omega \times X \to \mathcal{P}(X)$ a multifunction satisfying:*

1. *For each $x \in X$, the multifunction $\omega \mapsto F(\omega, x)$ is measurable*
2. *For each $\omega \in \Omega$, $F(\omega, \cdot)$ has closed, nonempty values*
3. *For some measurable function $k : \Omega \to \mathbb{R}^+$, the Lipschitz condition holds:*

$$H(F(\omega, x), F(\omega, y)) \leq k(\omega)\|x - y\| \quad \text{for all } x, y \in X \text{ and a.e. } \omega$$

*Then there exists a measurable selection $s : \Omega \to X$ such that*

$$s(\omega) \in F(\omega) \quad \text{for a.e. } \omega \in \Omega.$$

## Filipov's Selection Theorem

1. $F : \Omega \times X \to \mathcal{P}(Y)$ measurable

2. $F(\omega, \cdot)$ has closed values for all $\omega$

3. For $\sigma$-finite measure space $(\Omega, \mathcal{A}, \mu)$

$\Rightarrow$ **There exists a measurable selection** $s : \Omega \to Y$ **such that** $s(\omega) \in F(\omega)$ **for a.e.** $\omega \in \Omega$

# Applications of Filipov's Theorem

The Filipov Selection Theorem has important applications:

1. **Differential Inclusions**: Existence of solutions to

$$x'(t) \in F(t, x(t)), \quad x(0) = x_0$$

2. **Integral Inclusions**: Solving

$$x(t) = \lambda(t) + \int_0^t u(s, x(s))\, ds, \quad u(s, x) \in F(s, x)$$

3. **Optimal Control**: Finding measurable control functions in

$$\min_{u(t) \in F(t)} \int_0^T g(t, x(t), u(t))\, dt$$

4. **Variational Inequalities**: Proving solvability of structured variational problems

# Bochner Integral in Banach Spaces

## Definition (Strongly Measurable Function)

A function $u : \Omega \to X$ (where $X$ is a Banach space) is **strongly measurable** if there exists a sequence $\{u_n\}$ of simple functions such that

$$\|u_n(x) - u(x)\|_X \to 0 \quad \text{as } n \to \infty \text{ for a.e. } x \in \Omega.$$

## Definition (Bochner Integrable)

A strongly measurable function $u : \Omega \to X$ is **Bochner integrable** if there exists a sequence of simple functions $\{u_n\}$ such that

$$\int_\Omega \|u_n(\omega) - u(\omega)\|_X \, d\mu(\omega) \to 0 \quad \text{as } n \to \infty.$$

# Bochner Integral Definition

**Definition (Bochner Integral)**

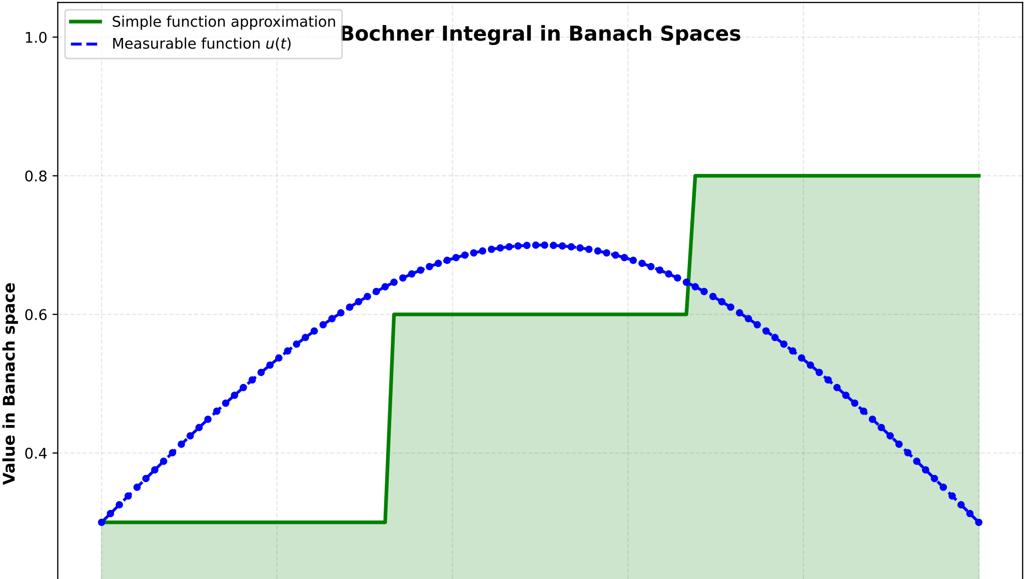
For a Bochner integrable function $u : \Omega \to X$, the **Bochner integral** is defined as

$$\int_E u \, d\mu = \lim_{n\to\infty} \int_E u_n \, d\mu,$$

where $\{u_n\}$ is a sequence of simple functions converging to $u$ in the sense above, and $E \in \mathcal{A}$.

The limit is independent of the choice of approximating simple functions.

# Bochner Integral Illustration



# Properties of Bochner Integral

## Theorem (Properties of Bochner Integral)

*Let $u, v : \Omega \to X$ be Bochner integrable. Then:*

1. **Linearity**: *For $a, b \in \mathbb{R}$,*
$$\int_E (au + bv)\, d\mu = a \int_E u\, d\mu + b \int_E v\, d\mu$$
2. **Countable Additivity**: *If $\{E_n\}$ is a countable partition of $E$,*
$$\int_E u\, d\mu = \sum_{n=1}^{\infty} \int_{E_n} u\, d\mu$$
3. **Norm estimate**:
$$\left\| \int_E u\, d\mu \right\|_X \leq \int_E \|u\|_X\, d\mu$$
4. **Dominated Convergence**: *If $\|u_n\| \leq g$ for a $\mu$-integrable $g$, then*
$$\int_E u_n\, d\mu \to \int_E u\, d\mu \quad \text{as } n \to \infty$$

# Existence of Solutions to Differential Inclusions

## Theorem (Existence for Differential Inclusions)

*Let $X$ be a separable Banach space, $[a, b]$ a compact interval, and $F : [a, b] \times X \to \mathcal{P}(X)$ a multifunction satisfying:*

1. *$F(t, \cdot)$ is measurable for each $x \in X$*
2. *$F(\cdot, x)$ has closed, convex values for each $x \in X$*
3. *$\|F(t, x)\| := \sup_{v \in F(t,x)} \|v\| \leq M(t)(1 + \|x\|)$ for some integrable $M : [a, b] \to \mathbb{R}^+$*

*Then for any $x_0 \in X$, there exists an absolutely continuous solution*

$$x : [a, b] \to X, \quad x(a) = x_0,$$

*to the differential inclusion*

$$\frac{dx}{dt} \in F(t, x(t)) \quad \text{for a.e. } t \in [a, b].$$

# Integral Inclusions

## Definition (Integral Inclusion)

An **integral inclusion** has the form

$$x(t) \in \lambda(t) + \int_0^t F(s, x(s))\, ds, \quad t \in [0, T],$$

where:

- $\lambda : [0, T] \to X$ is continuous
- $F : [0, T] \times X \to \mathcal{P}(X)$ is a measurable multifunction
- The integral is understood in the Bochner sense

## Theorem (Existence for Integral Inclusions)

*Under appropriate conditions on $F$ (measurability, closed convex values, growth conditions), there exists a continuous function $x : [0, T] \to X$ satisfying the integral inclusion.*

# Hammerstein Type Integral Equations I

**Definition (Nonconvex Hammerstein Integral Equation)**

A nonconvex Hammerstein-type integral equation has the form

$$x(t) = \lambda(t) + \int_0^T k(t,s)u(s,x(s))\,ds,$$

where $u(s,x) \in F(s,x)$ and $F$ is a nonconvex multifunction.

# Hammerstein Type Integral Equations II

**Theorem (Existence for Nonconvex Hammerstein Equations)**

*Assume:*

1. $(\Omega, \mathcal{A}, \mu)$ *is a finite measure space*
2. $F : \Omega \times X \to \mathcal{P}(X)$ *is measurable with closed, nonempty values*
3. *Growth condition:* $\|F(t, x)\| \leq c(t)(1 + \|x\|)$
4. $L(\cdot) \in L_1(\Omega)$, $\|k(\cdot, \cdot)\| \in L_\infty(\Omega \times \Omega)$

*Then there exists a solution* $x \in L_p(\Omega, X)$ *to the equation.*

# Hammerstein Type Integral Equations III

[Key Techniques] The proof typically uses:

- Filipov's selection theorem to choose measurable selections
- Bochner integral properties for integration of selections
- Fixed point theorems (Schauder, Banach) for existence
- Properties of measurable sets and countable unions

# Optimal Control Problems

## Definition (Optimal Control Problem)

Minimize the cost functional

$$J(u) = \int_0^T g(t, x(t), u(t))\, dt$$

subject to the constraints:

- $\frac{dx}{dt} = f(t, x, u)$ (system dynamics)
- $u(t) \in U(t)$ (control constraint)
- $x(0) = x_0$ (initial condition)

## Theorem (Existence of Optimal Control)

*Under appropriate measurability and convexity conditions on the dynamics and cost, there exists an optimal measurable control function $u^* : [0, T] \to U(t)$ that minimizes the cost functional.*

# Numerical Example: Solving a Differential Inclusion

## Example

Consider the differential inclusion

$$x'(t) \in [-1, 1 - \cos(x(t))], \quad x(0) = 0, \quad t \in [0, 1].$$

This models a system where the rate of change is constrained to lie in a given interval depending on the state.

**Key observations:**

- The multifunction $F(t, x) = [-1, 1 - \cos(x)]$ is measurable and has closed values
- Filipov's theorem guarantees a measurable selection $u(t) \in F(t, x(t))$
- The solution exists and can be computed numerically

# Python Implementation Outline

```
import numpy as np
from scipy.integrate import odeint

# Define the multifunction values
def F_bounds(t, x):
    lower = -1
    upper = 1 - np.cos(x)
    return lower, upper

# Numerical solution using Euler method with selection
def solve_differential_inclusion(x0=0, T=1.0, N=1000):
    dt = T / N
    t = np.linspace(0, T, N+1)
    x = np.zeros(N+1)
    x[0] = x0

    for i in range(N):
        # Choose a measurable selection (here: midpoint)
        lower, upper = F_bounds(t[i], x[i])
        u = (lower + upper) / 2.0
        x[i+1] = x[i] + u * dt

    return t, x

# Compute solution
t_vals, x_vals = solve_differential_inclusion()
```

# Results and Interpretation

**Solution characteristics:**

- The solution trajectory remains bounded: $|x(t)| \leq \|x_0\| + T$
- At each time $t$, the velocity is constrained: $x'(t) \in [-1, 1 - \cos(x(t))]$
- Multiple solutions exist (corresponding to different selections)
- All solutions are absolutely continuous

**Physical interpretation:**

- Lower bound $-1$: maximum negative velocity
- Upper bound $1 - \cos(x)$: state-dependent positive velocity
- When $x$ is small, upper bound is near 0 (slow growth)
- When $x$ is large, upper bound approaches 2 (fast growth)

# Summary

1. **Measure Theory**: Foundation for measurable sets and functions
2. **Multifunctions**: Set-valued maps that generalize standard functions
3. **Measurable Selections**: Measurable single-valued selections of multifunctions
4. **Filipov's Theorem**: Guarantees existence of measurable selections
5. **Integration**: Bochner integral extends classical integration to Banach spaces
6. **Applications**: Differential and integral inclusions, optimal control
7. **Numerical Methods**: Discrete approximations and implementations

# Key Takeaways

- **Importance of Measurability**: Essential for rigorous treatment of multifunctions
- **Powerful Existence Results**: Filipov's theorem enables solution existence proofs
- **Bochner Integration**: Proper framework for integrating functions into Banach spaces
- **Wide Applicability**: Results apply to differential inclusions, control problems, and variational inequalities
- **Theoretical and Practical**: Theory supports both existence proofs and numerical computations

# Further Reading


1. Pathak, H.K. (2018). *An Introduction to Nonlinear Analysis and Fixed Point Theory*. Springer.
2. Aubin, J.P., Cellina, A. (1984). *Differential Inclusions*. Springer.
3. Castain, M., Valadier, M. (1977). *Convex Analysis and Measurable Multifunctions*. Springer.
4. Kisielewicz, M. (1991). *Differential Inclusions and Optimal Control*. Kluwer.
5. Repovš, D., Semenov, P.V. (1998). *Continuous Selections of Multivalued Mappings*. Kluwer.

# Exercises

1. Prove that if $F$ and $G$ are measurable multifunctions with closed values, then so is $F \cap G$ (when $F(\omega) \cap G(\omega) \neq \emptyset$).
2. Let $f : \Omega \to X$ be a measurable function. Show that the multifunction $F : \Omega \to \mathcal{P}(X)$ defined by $F(\omega) = \{f(\omega)\}$ is measurable.
3. Construct an example of a multifunction that has measurable selections but is not itself measurable.
4. Prove the Bochner integral is countably additive: for disjoint sets $E_n$,

$$\int_{\cup E_n} u \, d\mu = \sum_n \int_{E_n} u \, d\mu.$$